



COURSE DESCRIPTION FORM

INSTITUTION FAST - National University of Computers and Emerging Sciences

PROGRAM (S) TO BE BS – CS

EVALUATED

A. Course Description

Course Title	Design and Analysis of Algorithms		
Course Code	CS2009	Credit Hours	3 + 0
Prerequisites by Course(s)	CS2001	Semester	Fall 2026
Assessment Instruments (With tentative weights)	Semester Work 10% (5 assignments) Project 10% (one project) Midterm 30% (Midterm1:15, Midterm2: 15) Final 50% (Comprehensive end of semester exam)		
Current Catalog Description	Main objective of course is to understand required skills and knowledge to design and analyze algorithms. Students will learn several algorithm design techniques and use of mathematical tools for empirical analysis of algorithms. Additionally, students will be able to learn effective problem-solving skills in computing through this course.		
Textbook (or Laboratory Manual for Laboratory Courses)	1. Thomas H. Cormen et al, Introduction to Algorithms, 4 th Edition		
Reference Material	1. Anany Levitin, Introduction to the design and analysis of algorithms, 3 rd Edition		
Course Goals	A. Course Learning Outcomes (CLOs)		
	CLO No.	Course Learning Outcomes	Bloom Taxonomy
	CLO-1	To apply acquired knowledge to solve computing problems complexities and proofs	C3 (Cognitive)
	CLO-2	To analyze complexities of different algorithms using asymptotic notations, complexity classes and standard complexity function	C4 (Cognitive)
			Tools
			A1, A2, A3, A4, A5, M1,M2, F
			A1, A2, A3, A4, A5, M1, M2, F

	CLO-3	To evaluate generic algorithmic solutions such as sorting, searching and graphs applied to real-world problems	C5 (Cognitive)	A3, A4, A5, M1, M2, F
	CLO-4	To construct and analyze real world problems solutions using different algorithms design techniques i.e. Brute Force, Divide and Conquer, Dynamic Programming, Greedy Algorithms.	C6 (Psychomotor)	Project
	Tool: A = Assignment, Q = Quiz, M = Midterm, F=Final, CEP = Complex Engineering Problem.			

B. Program Learning Outcomes		
PLO 1	Computing Knowledge	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems.
PLO 2	Problem Analysis	Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.
PLO 3	Design/Develop Solutions	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PLO 4	Investigation & Experimentation	Conduct investigation of complex computing problems using research-based knowledge and research-based methods
PLO 5	Modern Tool Usage	Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modeling for complex computing problems.
PLO 6	Society Responsibility	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems.
PLO 7	Environment and Sustainability	Understand and evaluate sustainability and impact of professional computing work in the solution of complex computing problems
PLO 8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of computing practice.
PLO 9	Individual and Team Work	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
PLO 10	Communication	Communicate effectively on complex computing activities with the computing community and with society at large.
PLO 11	Project Mgmt and Finance	Demonstrate knowledge and understanding of management principles and economic decision making and apply these to one's own work as a member or a team.

	PLO 12	Life Long Learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes.								
C. Relation between CLOs and PLOs (CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)											
		PLOs									
		1	2	3	4	5	6	7	8	9	10
CLOs	1	✓									
	2		✓								
	3					✓					
	4									✓	

Topics Covered in the Course (Tentative plan)	Topics	No. of Weeks	Contact Hours	CLO
	Algorithmic Thinking: Problem Formulation, Implementing the Definition, Algorithms Runtime Analysis and Basic Numbers and Vectors Arithmetic. Asymptotic Notations and Analysis: Runtime Analysis and Big Oh, Complexity Classes and Curse of Exponential Time, Relational Properties: Big Omega, Big Theta, Little Oh, Little Omega and Loop Invariant. (Project will be announced at the end of 2nd Week)	2	6	CLO-1
	Design Paradigm: Divide and Conquer, Mathematical Recurrence Relation: Substitution Method, Recurrence-Tree Method, Master's Method, Merge Sort, Finding Rank (Brute Force and Divide and Conquer Approach – Merge Sort), Karatsuba Integer Multiplication Algorithm, Counting Inversions (Brute Force and Divide and Conquer Approach, Examples, Application Theoretical Only), and Maximum Sum Subarray (Optional) (Assignment 01 will be released in 4th Week)	2	6	CLO-2

	Closest Pair of Points Algorithm and Time Complexity	1	3	CLO-2,3
	Geometric Algorithms: Introduction to Primitive Operations, Convex Hull and Closest Pair of Points Algorithm)	1	3	CLO-3
	Mid-I Examination	0.33	1	CLO-1,2,3
	Graphs and Trees: Basic Graph's and Tree's, Generic Greedy Interval Scheduling only, Greedy Interval Coloring and Huffman Coding Basics, Algorithm and Time Complexities, (Assignment 02 will be released in 8th Week)	2	9	CLO-2,3
	Dynamic Programming: Principle of Optimality, Optimal Substructure, Memoization and Tabulation Approach, Weighted Independent Sets, Weighted Interval Scheduling, Matrix Chain Multiplication, The Knapsack Problem. (Assignment 3 be released on 10th Week)	3	9	CLO-3
	Mid-II Examination	0.33	1	CLO-1,2,3,4
	Classes of Problems: P, NP, NP-Hard and NP Completeness, and Intractable Problems (Assignment 4 & 5 will be released 13th and 14th Week, respectively)	1	3	CLO-3
	Polynomial Time Reduction, Proving NP-Complete Problems and Approximation Algorithms	2	6	CLO-3
	Project Presentations	1	3	CLO-4
	Final	1	3	CLO-1,2,3,4
	Total	17	51(Including 3 hours final exam)	
Class Time Spent on (in credit hours)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues
	8	20	20	0



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Oral and Written Communications	Every student is required to submit all 5 assignments and potential oral project presentations.
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Instructor Name: *Anaum Hamid*

Instructor Signature: _____

Date: 10-08-2026